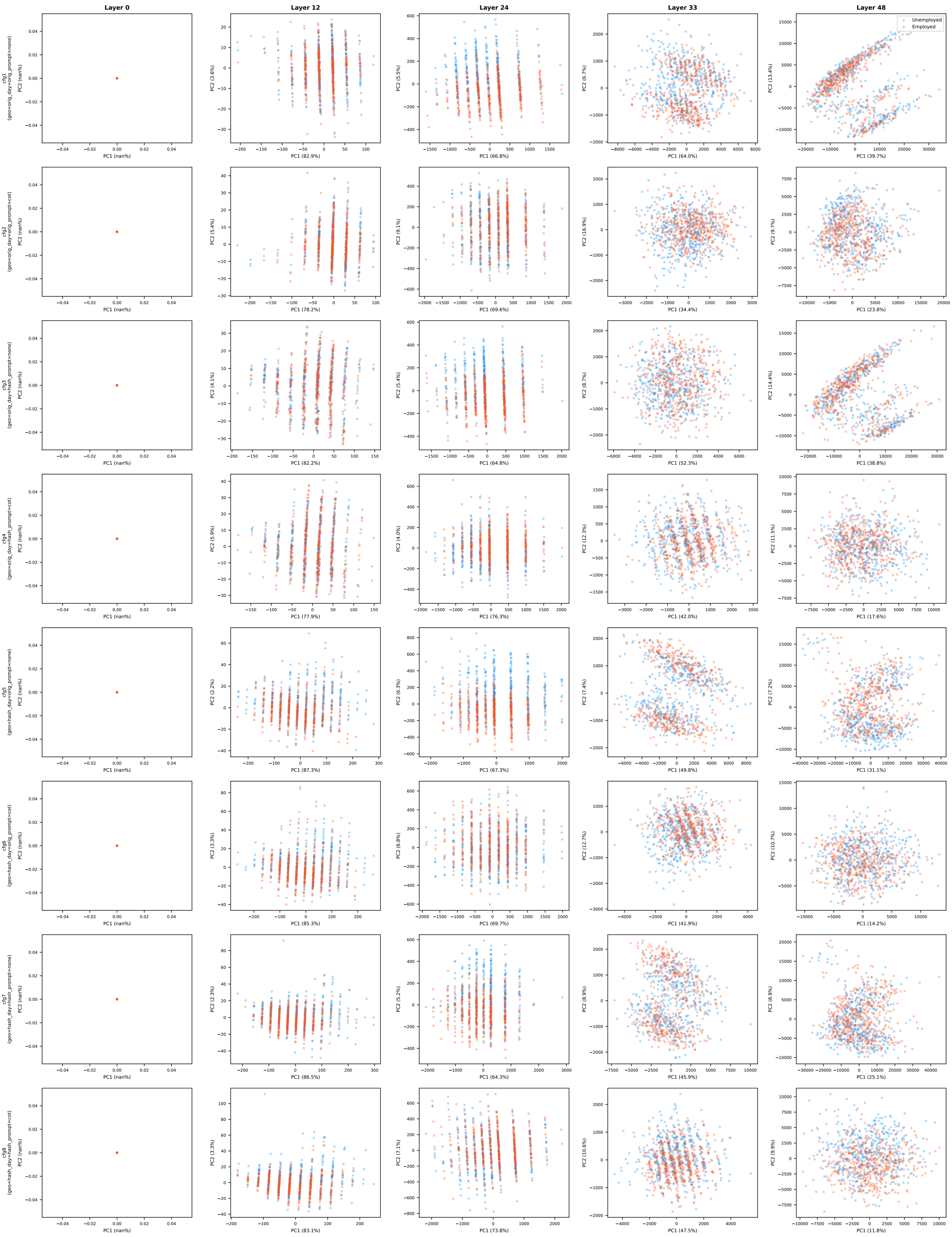
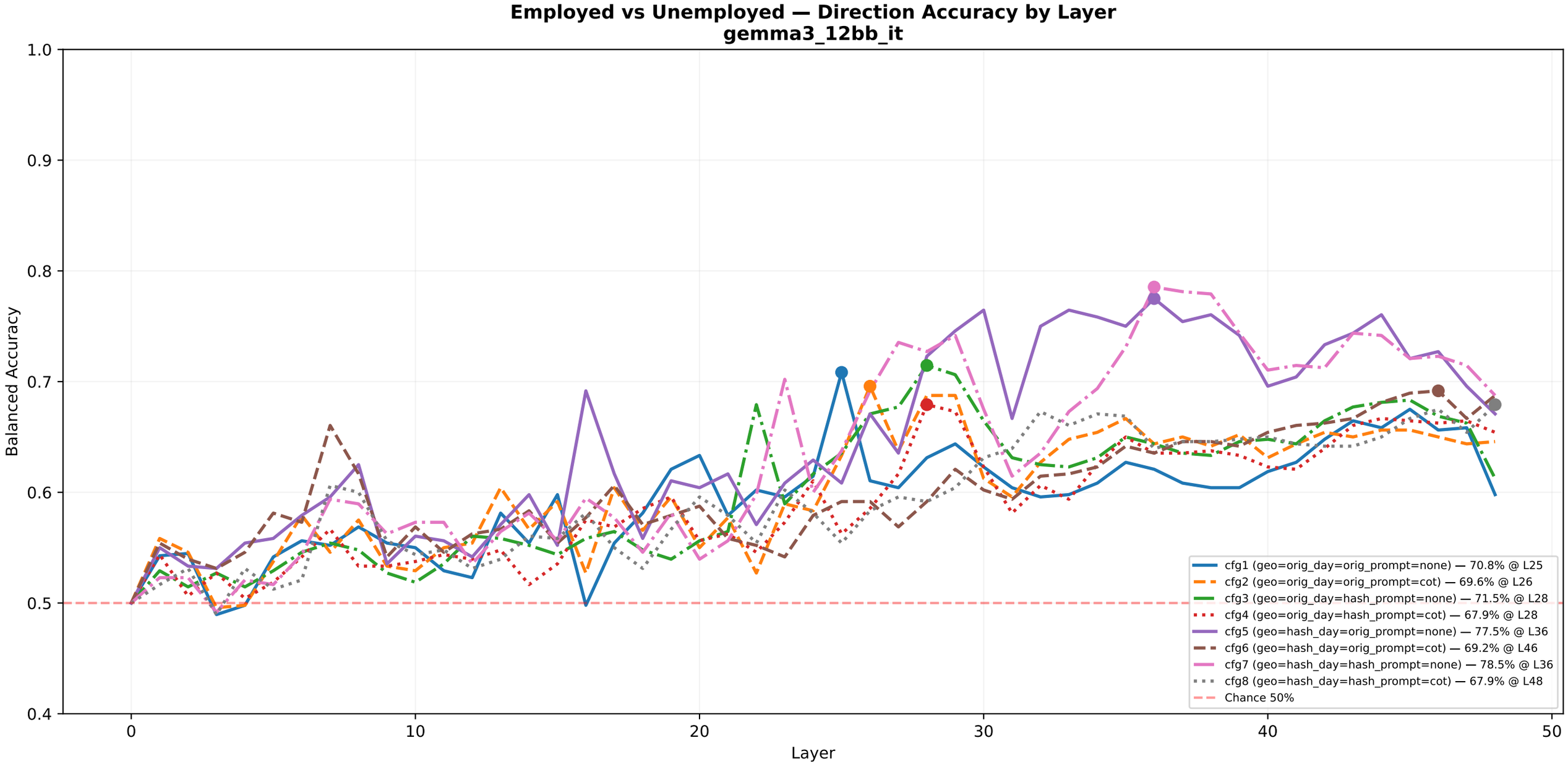
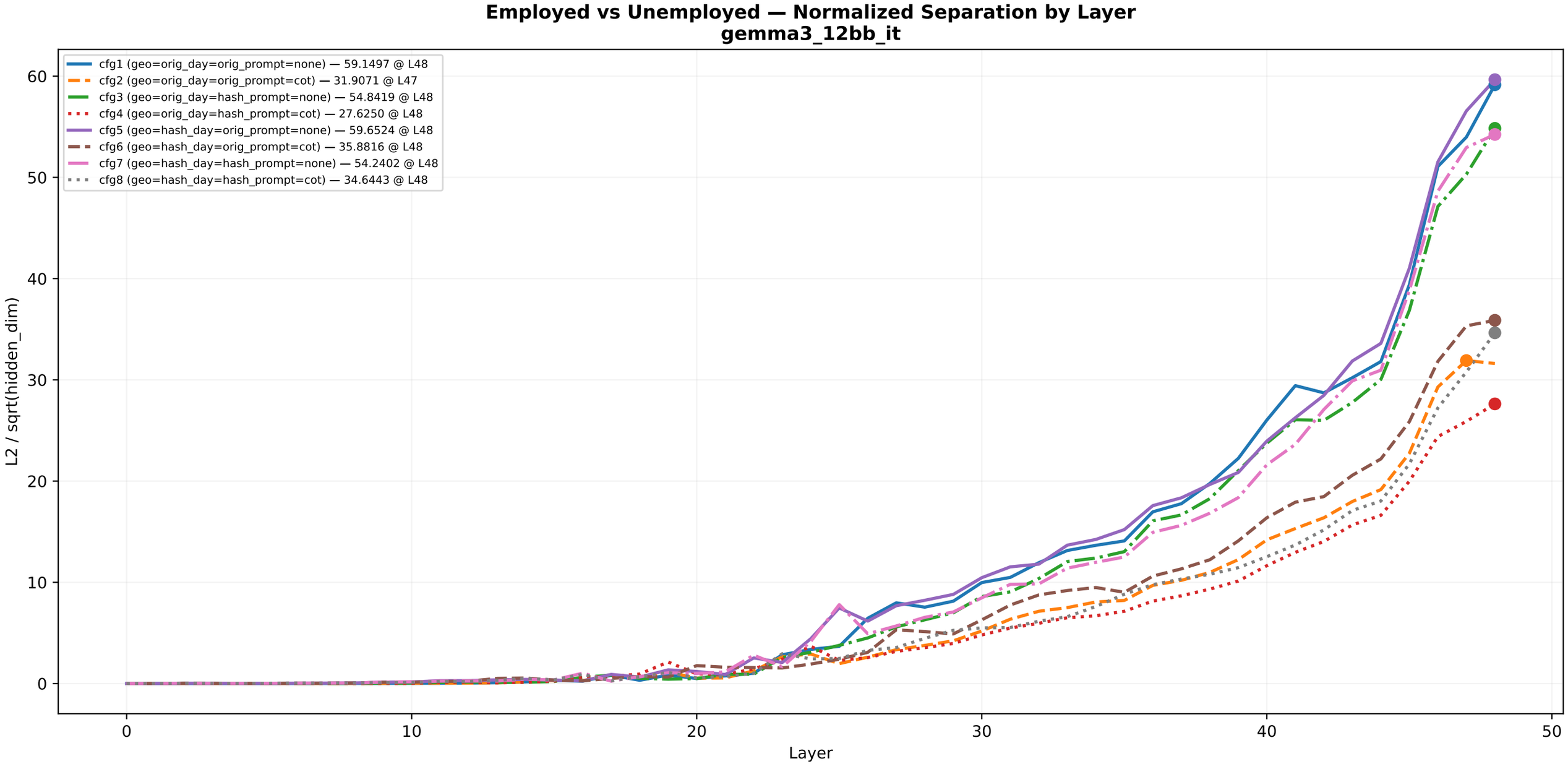
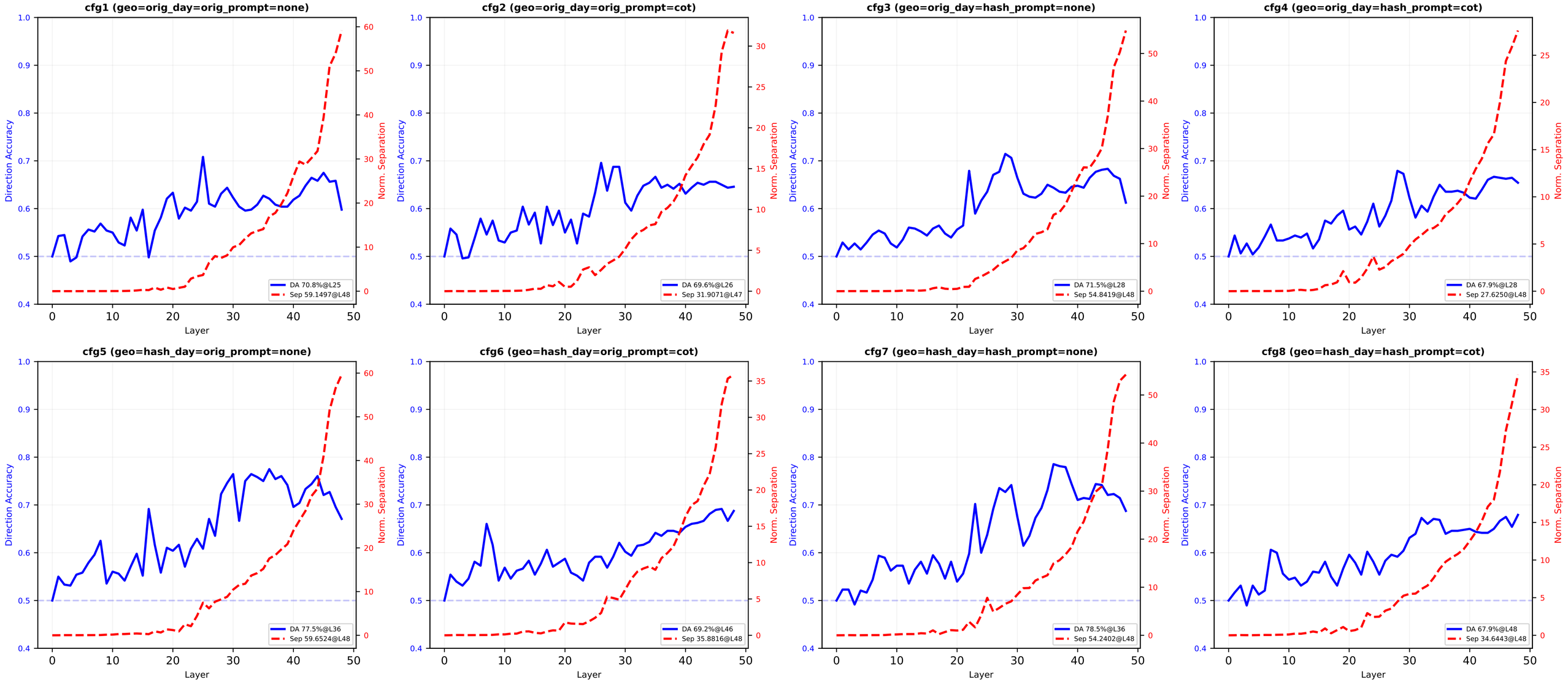
Employed vs Unemployed — PCA Scatter
gemma3_12bb_it | Layers: [0, 12, 24, 33, 48]




Employed vs Unemployed — Direction Accuracy by Layer
gemma3_12bb_it
Balanced Accuracy
Layer
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
10
20
30
40
50
cfg1 (geo=orig_day=orig_prompt=none) — 70.8% @ L25
cfg2 (geo=orig_day=orig_prompt=cot) — 69.6% @ L26
cfg3 (geo=orig_day=hash_prompt=none) — 71.5% @ L28
cfg4 (geo=orig_day=hash_prompt=cot) — 67.9% @ L28
cfg5 (geo=hash_day=orig_prompt=none) — 77.5% @ L36
cfg6 (geo=hash_day=orig_prompt=cot) — 69.2% @ L46
cfg7 (geo=hash_day=hash_prompt=none) — 78.5% @ L36
cfg8 (geo=hash_day=hash_prompt=cot) — 67.9% @ L48
Chance 50%


Employed vs Unemployed — Normalized Separation by Layer
gemma3_12bb_it
cfg1 (geo=orig_day=orig_prompt=none) — 59.1497 @ L48
cfg2 (geo=orig_day=orig_prompt=cot) — 31.9071 @ L47
cfg3 (geo=orig_day=hash_prompt=none) — 54.8419 @ L48
cfg4 (geo=orig_day=hash_prompt=cot) — 27.6250 @ L48
cfg5 (geo=hash_day=orig_prompt=none) — 59.6524 @ L48
cfg6 (geo=hash_day=orig_prompt=cot) — 35.8816 @ L48
cfg7 (geo=hash_day=hash_prompt=none) — 54.2402 @ L48
cfg8 (geo=hash_day=hash_prompt=cot) — 34.6443 @ L48
L2 / sqrt(hidden_dim)
Layer
0
10
20
30
40
50
60

Employed vs Unemployed — DA + Separation Per Config
gemma3_12bb_it



Employed vs Unemployed — Summary

Employed vs Unemployed — Geometry Summary | gemma3_12bb_it

cfg	label	DA_L	DA%	Sep_L	Sep	N
1	geo=orig_day=orig_prompt=none	L 25	70.8%	L 48	59.14975	960
2	geo=orig_day=orig_prompt=cot	L 26	69.6%	L 47	31.90706	960
3	geo=orig_day=hash_prompt=none	L 28	71.5%	L 48	54.84191	960
4	geo=orig_day=hash_prompt=cot	L 28	67.9%	L 48	27.62503	960
5	geo=hash_day=orig_prompt=none	L 36	77.5%	L 48	59.65237	960
6	geo=hash_day=orig_prompt=cot	L 46	69.2%	L 48	35.88164	960
7	geo=hash_day=hash_prompt=none	L 36	78.5%	L 48	54.24019	960
8	geo=hash_day=hash_prompt=cot	L 48	67.9%	L 48	34.64430	960

DA = Direction Accuracy (balanced acc of median-split projection)
Sep = Normalized Separation (L2(centroid diff) / sqrt(hidden_dim))
PCA layers: [0, 12, 24, 33, 48]